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EPIDEMIOLOGY

KENYA MEDICAL TRAINING COLLEGE-SIAYA
DEPARTMENT OF HEALTH RECORDS AND INFORMATION

By,
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MODULE TITLE: EPIDEMIOLOGY

TIME: 60 HOURS

MODULE COMPETENCE: The course is designed to equip the learners with the knowledge and skills of disease determinants and distribution for effective prevention and control of diseases

Expected learning outcomes

By the end of the unit the learner should be able to:

1. Apply the concepts of epidemiology in health care service
2. Describe the distribution and determinant of diseases
3. Calculate, interpret, and apply basic measure of morbidity, mortality and association
4. Describe the major types of epidemiologic research
5. Carry out epidemiologic studies
6. Demonstrate the principles of disease measurements

UNIT CONTENTS

Basic concepts of epidemiology and disease surveillance, definition of epidemiology and its origin, health, disease and risk factors, community health diagnosis, natural history of disease, ecological causes of disease, subject matter of epidemiology, scope of epidemiology.

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Course Outline

UNIT ONE: INTRODUCTION TO EPIDEMIOLOGY

- Definition of Epidemiology and its Origin
- Terms used in epidemiology
- Determinants of disease
- Occurrences of diseases
- Natural history of disease
- Level of disease prevention
- Ecological approach to disease causation
- Scope of epidemiology
- Uses and purposes of epidemiology

UNIT TWO: MEASURES OF MORBIDITY, MORTALITY AND ASSOCIATION

- Rates, ratios and proportions;
- Measures of morbidity (incidence and prevalence);
- Measures of mortality;
- Relative risk, absolute risk, attributable risk, odds ratio
- Causation; causal and non-causal relationships
- Sources of epidemiological data; census, vital statistics/registration systems, health service records (health information sources), morbidity and mortality surveys, demographic and health survey

UNIT THREE: DYNAMICS OF DISEASE TRANSMISSION

- Communicable disease epidemiology
- Chain of disease transmission
- Methods of controlling communicable diseases
- Vaccine preventable diseases
- Expanded Program on Immunisation (EPI)

UNIT FOUR: DISEASE SURVEILLANCE AND OUTBREAK INVESTIGATIONS

- Definitions
- Types of epidemics
- Detection of epidemics
- Investigation of epidemics/epidemics
- Control of epidemics
- Definitions, types and purposes of surveillance
- Components of surveillance
- Sources of surveillance data
- Steps in surveillance
- Strategy in disease surveillance
- Integrated Disease Surveillance and Response (IDSR)

UNIT FIVE: EPIDEMIOLOGICAL DATA AND STUDY DESIGNS

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- Sources of epidemiological data
- Experimental studies, observational studies, descriptive studies, analytical studies, cross-sectional studies, longitudinal studies, case-control studies

UNIT SIX: PRINCIPLES OF DISEASE MEASUREMENTS

- Reliability, observer variation, variability, test bias and confounding, error and systematic error, repeatability, validity (content, concurrent, predictive and construct), sensitivity and specificity

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UNIT ONE: INTRODUCTION TO EPIDEMIOLOGY

Key concepts in epidemiology

Some basic terms

HEALTH- is defined by WHO as a state of complete physical, mental and social well being and not merely the absence of disease or infirmity.

DISEASE- is a physiological/ psychological dysfunction.

ILLNESS- is a subjective state of a person who feels aware of not being well.

RISK FACTOR- is a factor whose presence is associated with an increased probability that the disease will develop later for example the habit of smoking is a risk factor for lung cancer. Risk factors may be susceptible to change, such as smoking habits, or may not be subject to change e.g. sex, family history. The identification of such factors is useful in order to identify persons who deserve close medical supervision.

VEHICLE - An inanimate intermediary in the indirect transmission of an agent that carries the agent from a reservoir to a susceptible host.

ZOONOSES An infectious disease that is transmissible under normal conditions from animals to humans.

COMMUNITY- it refers to a group of people who share something in common. Examples of common denominator may be neighborhood, workplace, race, religion or social activity.

COMMUNITY MEDICINE- is also referred to as population medicine, preventive medicine, social medicine or more traditionally as public health. Unlike clinical medicine which focuses largely on medical care of individuals, community medicine is concerned with the study of health and disease in human populations.

AGENT A factor, such as a microorganism, chemical substance, or form of radiation, whose presence, excessive presence, or (in deficiency diseases) relative absence is essential for the occurrence of a disease.

RESERVOIR The habitat, in which an infectious agent normally lives, grows and multiplies; reservoirs include human reservoirs, animals' reservoirs, and environmental reservoirs.

NATURAL HISTORY OF DISEASE the process of changes or pathogenesis taking place from

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Epidemiology Definition:

WHO Definition:

Epidemiology is the study of the distribution and determinants of health related states or events in a specified population and the application of this study to control health problems.

Other definition

It can also be the study of distribution and determinant of health related conditions and events in population and application of these studies.

Components of the Definition

Distribution:

Epidemiology is concerned with the **frequency** and **pattern** of health events in a population. **Frequency** includes not only the number of such events in a population, but also the rate or risk of disease in the population. The rate (number of events divided by size of the population) is critical to epidemiologists because it allows valid comparisons across different populations.

Patterns refer to the occurrence of health-related events by **time, place and person characteristics.**

Time characteristics include annual occurrence, seasonal occurrence, and daily or even hourly occurrence during an epidemic

Place characteristics include geographic variation, urban-rural differences, and location of work sites or schools

Person characteristics include demographic factors such as age, race, sex, marital status and socio economic status, as well as behaviors and environmental exposures

This characterization of the distribution of health-related states or events is one broad aspect of epidemiology called **descriptive epidemiology**. Descriptive epidemiology provides the (What Who When and Where) of health related events.

Determinants:

Epidemiology is also used to search for causes and other factors that influence the occurrence of health-related events

Analytic epidemiology attempts to provide the **Why and How** of such events by **comparing groups with different rates of disease occurrence and with differences in demographic** characteristics, genetic or immunologic make-up, behaviors, environmental exposures, and other

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so-called potential risk factors. Under ideal circumstances, epidemiologic findings provide sufficient evidence to direct swift and effective public health control and prevention measures.

Health-related states or events

Originally, epidemiology was concerned with **epidemics of communicable diseases.** Then epidemiology was extended to **endemic communicable diseases** and non-communicable infectious diseases. More recently, epidemiologic methods have been applied to chronic diseases, injuries, birth defects, maternal-child health, occupational health and environmental health. Now, even behaviors related to health and well-being (amount of exercise, seat-belt use, etc) are recognized as valid subjects for applying epidemiologic methods. "Disease" is a term commonly used to refer to the range of health-related states or events

Specified populations

Although epidemiologists and physicians in clinical practice are both concerned with disease and the control of disease, they differ greatly in how they view "**the patient.**" Clinicians are concerned with the health of an individual; epidemiologists are concerned with the collective health of the people in a community or other areas. When faced with a patient with diarrheal disease, for example, the clinician and the epidemiologist have different responsibilities. Although both are interested in establishing the correct diagnosis, the clinician usually focuses on treating and caring for the individual. The epidemiologist focuses on the exposure (action or source that caused the illness), the number of other persons who may have been similarly exposed, the potential for further spread in the community and interventions to prevent additional cases or recurrences.

Application

Epidemiology is more than "the study of." As a discipline within public health, epidemiology provides data for directing public health action. However, using epidemiologic data is an art as well as a science. Consider again the medical model used above: To treat a patient, a clinician must call upon experience and creativity as well as scientific knowledge. Similarly, an epidemiologist uses the scientific methods of descriptive and analytic epidemiology in "diagnosing" the health of a community, but also must call upon experience and creativity when planning how to control and prevent disease in the community.

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Historical Evolution of Epidemiology

Although epidemiology as a subject has blossomed since World War II, epidemiologic thinking has been traced from Hippocrates through John Graunt, William Farr, John Snow and others. The contributions of some of these early and more recent thinkers are;

i. Hippocrates (Circa 400 B.C)

Hippocrates attempted to explain disease occurrence from a rational rather than a supernatural viewpoint. In his essay entitled "On Airs, Waters, and Places," Hippocrates suggested that environmental and host factors such as behaviors might influence the development of disease.

ii. 1662

Another early contributor to epidemiology was John Graunt, a London haberdasher (businessman) and councilman who published a landmark analysis of mortality data in 1662. This publication was the first to quantify patterns of birth, death and disease occurrence, noting disparities between males and females, high infant mortality, urban/rural differences, and seasonal variations.

iii. 1800

No one built upon Graunt's work until mid-1800s. William Farr built upon Graunt's work by systematically collecting and analyzing Britain's mortality statistics. Farr is considered the father of modern vital statistics and surveillance because he developed many of the basic practices used today in vital statistics and disease classification. He extended the epidemiologic analysis of morbidity and mortality data, looking at the effects of marital status, occupation and altitude. He concentrated his efforts on collecting vital statistics, assembling and evaluating those data and reporting to responsible health authorities and the general public. He also developed many epidemiologic concepts and techniques still in use today.

iv. 1854

Meanwhile, an anesthesiologist named John Snow was conducting a series of investigations in London that later earned him the title "the father of field epidemiology." Twenty years before the development of the microscope, Snow conducted studies of cholera outbreaks both to discover the cause of disease and to prevent its recurrence. Because his work classically illustrates the sequence from descriptive epidemiology to hypothesis generation to hypothesis testing (analytic epidemiology) to application, we will consider two of his efforts in detail.

Snow conducted his classic study in 1854 when an epidemic of cholera developed in the Golden Square of London. He began his investigation by determining where in this area persons

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with cholera lived and worked. He then used this information to map the distribution of cases on what epidemiologists call a spot map.

Because Snow believed that water was a source of infection for cholera, he marked the location of water pumps on his spot map, and then looked for a relationship between the distribution of cholera case households and the location of pumps. He noticed that more case households clustered around Pump A, the Broad Street pump, than around Pump B or C, and he concluded that the Broad Street pump was the most likely source of infection. Questioning residents who lived near the other pumps, he found that they avoided Pump B because it was grossly contaminated, and that Pump C was located too inconveniently for most residents of the Golden Square area. From this information, it appeared to Snow that the Broad Street pump was probably the primary source of water for most persons with cholera in the Golden Square area. He realized, however, that it was too soon to draw that conclusion because the map showed no cholera cases in a two-block area to the east of the Broad Street pump. Perhaps no one lived in that area. Or perhaps the residents were somehow protected.

Upon investigating, Snow found that a brewery was located there and that it had a deep well on the premises where brewery workers, who also lived in the area, got their water. In addition, the brewery allotted workers a daily quota of malt liquor. Access to these uncontaminated rations could explain why none of the brewery's employees contracted cholera.

To confirm that the Broad Street pump was the source of the epidemic, Snow gathered information on where persons with cholera had obtained their water. Consumption of water from the Broad Street pump was the one common factor among the cholera patients. According to legend, Snow removed the handle of that pump and aborted the outbreak.

Snow's second major contribution involved another investigation of the same outbreak of cholera that occurred in London in 1854. In a London epidemic in 1849, Snow had noted that districts with the highest mortalities had water supplied by two companies: the Lambeth Company and the Southwark and Vauxhall Company. At that time, both companies obtained water from the Thames River, at intake points that were below London. In 1852, the Lambeth Company moved their water works to above London, thus obtaining water that was free of London sewage. When cholera returned to London in 1853, Snow realized the Lambeth Company's relocation of its intake point would allow him to compare districts that were supplied with water from above London with districts that received water from below London. The table below shows what Snow found when he made that comparison for cholera mortality over a 7-week period during the summer of 1854.

Table showing mortality from cholera in the districts of London supplied by the Southwark and Vauxhall and the Lambeth companies, July 9 to August 26, 1854

Districts with water supplied by:	**Population (1851 census)**	**Number of deaths from cholera**	**Cholera death rate per 1,000 population**
Southwark and Vauxhall only	167,654	844	5.0
Lambeth only	19,133	18	0.9
Both companies	300,149	652	2.2

The data in the above table show that the risk of death from cholera was more than 5 times higher in districts served only by the Southwark and Vauxhall Company than in those served

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only by the Lambeth Company. Interestingly, the mortality rate in districts supplied by both companies fell between the rates for districts served exclusively by either company. These data were consistent with the hypothesis that water obtained from the Thames below London was a source of cholera. Alternatively, the populations supplied by the two companies may have differed on a number of other factors which affected their risk of cholera.

To test his water supply hypothesis, Snow focused on the districts served by both companies, because the households within a district were generally comparable except for water supply company. In these districts, Snow identified the water supply company for every house in which a death from cholera had occurred during the 7-week period. Table 1.2 shows his findings.

This further study added support to Snow's hypothesis, and demonstrates the sequence of steps used today to investigate outbreaks of disease. Based on a characterization of the cases and population at risk by time, place, and person, Snow developed a testable hypothesis. He then tested this hypothesis with a more rigorously designed study, ensuring that the groups to be compared were comparable. After this study, efforts to control the epidemic were directed at changing the location of the water intake of the Southwark and Vauxhall Company to avoid sources of contamination. Thus, with no knowledge of the existence of microorganisms, Snow demonstrated through epidemiologic studies that water could serve as a vehicle for transmitting cholera and that epidemiologic information could be used to direct prompt and appropriate public health action.

v. 19th and 21st centuries

Branches of Epidemiology

Descriptive epidemiology (Distribution)

It describes the occurrence of a disease and other health-related characteristics in human population in terms of:

1. Person (the who; age, sex and social class etc.)
2. Place (the where, the geographical location, residence etc.)
3. Time (the when- season, time of day/ week etc)

Analytical epidemiology (determinant)

It investigates the association between a given disease or health status and possible causative or protective factors e.g. ask- why does health related outcomes occur?

Epidemiology forms the research arm of public health providing the scientific basis upon which public health policy decisions are made.

It provides methodological research skills in order to manage public health problems in a more effective way.

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Public health

It is defined in (last's dictionary of epidemiology) as one of the efforts organized by society to protect, promote and restore the health of a population. It is a combination of sciences, skills and beliefs that are directed to the maintenance and improvement of the health of the people through collective and social action.

Origin of public health – public health has its origin in attempt to control epidemics through the segregation and quarantine of affected individuals during the industrial revolution.

NB: It's during the industrial revolution that methods of environmental control including occupational health were developed and sanitary reforms promoted.

Important distinction in the use of the term epidemiology

i) **Epidemiology of a particular disease**

This describes the distribution and determinants of that disease e.g. epidemiology of HIV infections demonstrates that young people aged 16-30 years have the highest incidence rates of HIV infection.

Women have a three field higher risk of HIV infections and that approximately 25% of children born to HIV positive women are infected with HIV.

ii) **Epidemiology (as a discipline or subject)**

This describes particular methodological tools/ approaches used to gather information about the prevalence distribution, determinants and course of any condition.

Epidemiology is not just the study of diseases; it covers every aspect of health care. This might be the provision of health services, utilization of staff, even the deployment of limited resources into particular areas or at particular times of the year.

As a health professional, if you are posted at to a rural area you will find that you have so many sick patients that you are unable to give them all the kind of attention that a clinician would give a patient in order to formulate the correct course of treatment.

However, when you look at the whole community you will see that there are health problems which are shared by everyone e.g. malaria might be affecting the whole community, whereas diarrhea, measles and malnutrition are most prevalent among children. Many babies are being

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born but a few of these births are in hospitals so that your antenatal and postnatal care is limited to a small proportion of women only.

Questions

1. How can you plan health services for these mothers?
2. What is your strategy for vaccination for their children?
3. How can you control the disease common among the people?

Epidemiology is the study of all these factors.

An example of epidemiological study (epidemiology of snake bite).

A doctor in an isolated mission hospital in northern Nigeria realized that the farmers in the savanna area served by the hospital were often bitten by snakes at the end of the dry season and beginning of the rains.

They were clearing, hoeing and planting during those very months and so he collected data about admission for snake bites month by month.

The snake *echiscarinatus* the carpet viper is a terrible threat to savanna farmers.

The Doctor showed clearly the seasonal fluctuations in the incidence and mortality of "*E. Carinutus*" bites. Obviously these seasonal data showed that in his management at the hospital"*Echis*" anti-venom must be available abundantly during the peak season. The data also showed that he must inform the primary health care workers of the needs to be vigilant during the peak season and to refer cases then. This study shows the need for a careful collection of facts, disciplined record keeping and the value of data collected over a long period.

Therefore, epidemiology is all about the collection of information.

Clinical approach as an example of the collection of data in a country

You should approach a patient and consider him/ her thus Why should this person from this place fall ill in this way at this time?

The approach should be the example while collecting data in a community because epidemiology is about information.

The collection of information in epidemiology

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Epidemiology is about information needed for health planning, supervision and evaluation of the health promotion and disease control activities. The key components of the data needed can be approached through a series of questions.

i. Who is affected referring to age, sex, social class, ethnic group occupation, heredity and personal habits (these are person factors).

ii. Where did it happen, in relation to place of residence, geographical distribution and place of exposure? (These are the place factors).

iii. When did it happen in terms of the month, season or year? (These are time factors).

iv. What is the disease or condition, its clinical manifestations and diagnosis?

v. How did the disease occur, in relation to the interplay of the specific population, vector, source of infection, susceptible groups and other contributing factors?

vi. Why did it occur in terms of the reasons for the disease outbreak (e.g. broken down of sanitary services)?

vii. What now?- the most important question- what action is now to be taken as a result of the information gained.

The denominator

The most important idea in epidemiology is that of the denominator. It is vital to know about the apparently healthy non- patient as well as the obviously ill patient and the population at risk of developing a disease problem as well as those who presently have it. The idea of the denominator is key to keeping the needed information for health planning, supervision of activities and the evaluation of health promotion and disease control activities.

Who is an epidemiologist?

Epidemiologist is the person who:-

i. Studies the etiology, diagnosis and prevention of diseases in the community.

ii. Measures the frequency of disease in a specific population.

iii. Interprets the finding and usually recommends practicable preventive measures.

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iv. Instead of thermometer or stethoscope, use figures, calculators or computers from which to make his analysis and derive his deductions.

v. Counts cases of specific disease in relation to a census of the population at risk and the environmental factors to which they may be exposed.

Scope of epidemiology

Historically, epidemiology was largely concerned with infectious diseases and mainly epidemics. This scope of epidemiology has expanded to include both infectious and non-infectious diseases such as nutritional deficiency states, mental disorders, hypertension, accidents, cancer, congenital anomalies and degenerative diseases.

The use of epidemiology (purpose)

You may be able to make use of epidemiological information in the following ways:

i. Diagnostic purpose

Epidemiological studies provide data for community diagnosis which encompasses the identification and detailed analysis of needs, demands, priorities and resources of a defined community.

This would serve as a basis for the planning and implementation of health intervention measures (this will usually be a cross sectional study as observations are made at specific time.)

ii. Identification of determinants of diseases

One of the most important purposes of epidemiological studies is the identification of determinants of diseases whose manipulation could lead to prevention and control ofdiseases. Hypothesis about disease causation may arise directly out of simple epidemiological description of diseases in terms of the kind of people affected by its (age, sex, occupation etc.) its geographical distribution and variation in its frequency of occurrence at different times e.g. pellagra and maize diet.

Surveillance

To produce enhanced surveillance for specific diseases e.g. cholera in order to be able to act quickly and to cut short any outbreak or pertussis (whooping cough). Surveillance is a useful monitor for the coverage achieved by an epidemiology programme.

iii. Investigation of an outbreak

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To investigate an outbreak of a communicable disease, analyze the reason for it, plan a feasible remedy and carry it out and monitor the effects of the remedy on the outbreak.

iv. Observation of the natural history of diseases.

Knowledge of diseases natural history is essential to make prognosis or the likely outcome of a patient illness. This prognosis is the basis for rational decisions about therapy.

v. Classification of a disease

The epidemiological characteristics of a disease are an integral part of its basic description by means of which it is defined

and recognized e.g.

Gonorrhea classified as a STD based on its mode of transmission.

Shigellosis is grouped under diarrheal diseases based on its clinical manifestation of diarrhea and its fecal- oral route of transmission.

vi. Planning

To plan effective health services. This particularly derived from community diagnosis for which effective services. Interventions and remedies all depend on accurate community data.

vii. Evaluation of methods of disease control

Any programme designed to prevent and control a disease must be accompanied by methods for accessing whether the measures are effective in reducing the frequency of the diseases. If the study is not effective then necessary alterations are done accordingly.e.g.

The effectiveness of health education on infant feeding to mothers attending anti-natal is evaluated by measuring decline in the incidence of marasmus and kwashiorkor among children and the change in knowledge, attitude and behavior of mothers. (This will be a longitudinal study as it involves the collection of information over a period of time and monitoring the change of health in a community.)

COMMUNITY DIAGNOSIS

It is the process of identification and description of the most important health problems of a given community.

Community diagnosis provides for a basis for decision on

- The need for intervention.

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- Type of intervention.
- Target group at whom intervention should be directed.
- Provides baseline data for later intervention.

Some of the ways workers would make community diagnosis are:

- Discuss with the community leaders and health workers.
- Attending community gathering to identify community health problems.
- Survey available health records (daily clinical records, monthly and annual reports of health institution.)
- Field survey- conduct health interviews. Do physical examination and collect specimen for laboratory tests (sample surveys).
- Analysis of data collected and compilation of report in order to reach community diagnosis.

COMMUNITY MEDICINE- Involves discussion with the community + health records survey + field survey ⟶ community diagnosis ⟶ intervention ⟶ monitoring and evaluation.

Natural history of diseases

Natural history refers to the systematic description of the course of a disease overtime unaffected by treatment. Each disease has its own life history. There are four main stages in the natural history of diseases.

1. **Stage of susceptibility**

This is a stage in which the disease has not developed; the ground work has been laid by the presence of factors that favors its occurrence. Example

- Acute or chronic alcoholism increases susceptibility to liver cirrhosis, pneumonia.
- High serum cholesterol level increases risk of coronary heart diseases.
- An unvaccinated child is susceptible to measles.

2. **Stage of pre- symptomatic diseases(Sub- clinical stage)**

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In this stage, there is no manifest disease, that means no sign or symptoms are detected. The patient doesn't know that he has any disease. In some infectious disease, the bacterium, virus or parasite may enter and multiply in the body without causing symptoms or signs of disease example:

- The ova of intestinal parasite in the stool of apparently healthy children.
- Antibodies for polio-viruses are present in many adults but less than 1% have had symptoms or signs of diseases.

3. Stage of clinical

In this stage, one person has symptoms and signs of disease. Some diseases are short and mild that almost everyone recovers quickly e.g. common cold and chicken pox.

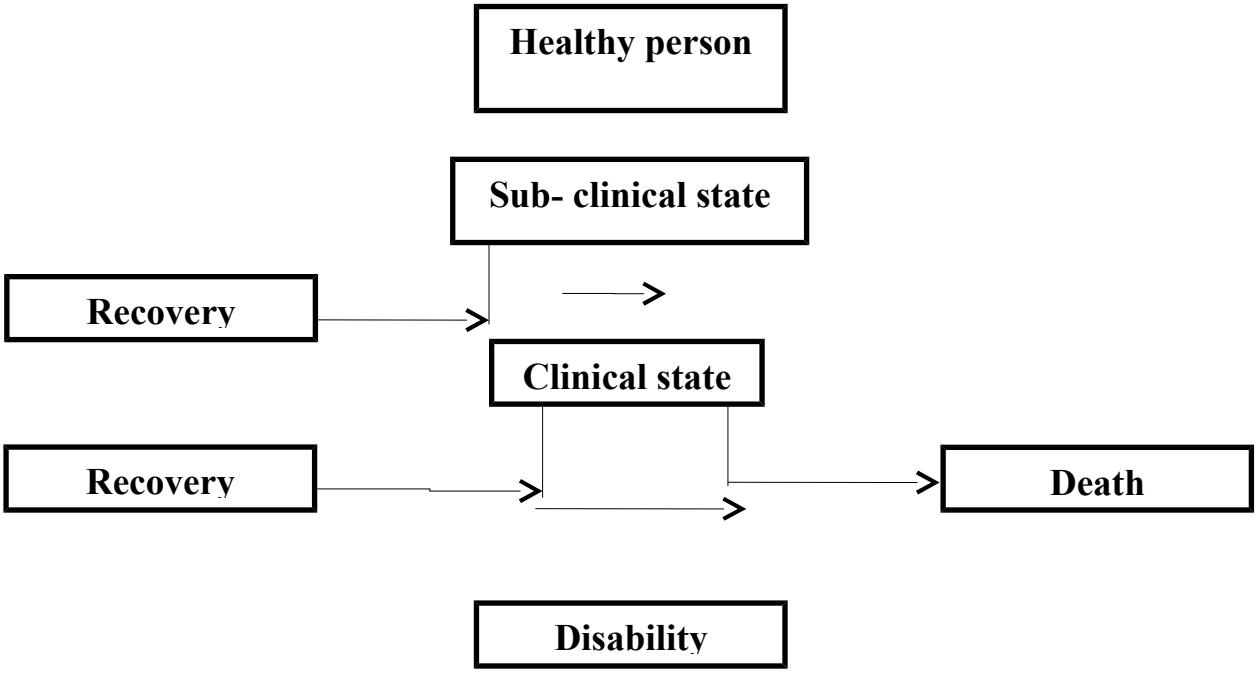
Others are very serious leading to complications and death e.g. babies leads to death and poliomyelitis can lead to paralysis which causes which causes permanent disability or death.

4. Stage of disability

Some diseases take this course and then solve completely either spontaneously or under the influence of therapy. However, there are a number of conditions which give rise to residual defects on short or long durations, leaving the person disabled to a greater or lesser extent.

In community surveys, disability usually means any limitation of a person's activity including his role as a part wage earner and member of his community.

Diagrammatic presentation of the stages



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DISEASE PREVENTION

Definition; actions aimed at eradicating, eliminating, or minimizing the impact of a disease and disability

LEVELS OF PREVENTION

The important tasks of health workers are:-

- ✓ Preventing people from getting sick.
- ✓ Preventing the sick people from getting worse.
- ✓ Or becoming disabled or chronically sick or dying.

There are three important ways that health workers can prevent the development of diseases.

Primordial prevention

It consists of actions and measures that inhibit the emergence of risk factors in the form of environmental, economic, social and behavioral conditions and cultural patterns of living e.t.c

It is the prevention of the emergence or development of risk factors in countries or population gropus in which they have not yet appeared eg *discouraging children from adopting harmful lifestyles to avoid development of diseases that occur during adulthood*

The main intervention is through individual and mass education

Primary prevention

This is the level where intervention is aimed at healthy people, individuals or groups. Healthy people are prevented from becoming sick.

Measures are taken to promote optimum health or provide specific protection of targets groups against diseases and injury.

In order to carry effective primary prevention we must ask why healthy people become sick and also know who is "most at risk" of getting the disease. Some of the methods of primary prevention are:-

1. Immunization.
2. Provision/supply of safe water

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3. Provision of adequate housing
4. Safe disposal of human excreta
5. Control of vectors of diseases
6. Protection against accidents and occupational hazards
7. Good nutrition, adequate clothing rest and recreation
8. Changing people's behavior through health education

Secondary prevention

This involves detecting people (in the subclinical stage) who already have the disease as early as possible in order to cure the patient and prevent chronic disease and disability.

Measures are aimed at diagnosis and prompt treatment to prevent the development of clinical disease and complications from the disease. This can be done by carrying a household survey or by telling people to visit the clinic to be examined by:

- Using laboratory tests to detect sub- clinical diseases
- Using clinical examination and lab tests to detect early clinical disease.

Tertiary prevention

This is the prevention of more disability and death. Measures at this level involve the treatment of the diseases (condition in its later stages and rehabilitation to optimize function thus preventing/ minimizing, impairment and disability.).

In a patient who cannot be cured, tertiary prevention has two aims;

- ✓ Treatment to prevent more disability and death
- ✓ New training and special education to help the patient to return to some useful work and life in the community (occupational therapy)

CAUSATION OF DISEASE

Causes of disease are rarely a single entity. Several factors often need to be present before the disease process occurs and becomes clinically evident. These factors are referred to as "risk factors", agents, causes or causative factors.

These are two central concepts in the causation of disease namely:-

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1. A necessary cause.

2. A sufficient cause.

Necessary cause- this is a combination of risk factor that must be a single risk factor. For example malaria parasite (plasmodium) is present.

If disease does not develop without the factor being present, then we term the causative factor **"necessary"**

Sufficient cause- Will invariability (always) lead to disease if it is present. A sufficient cause in distinct from a necessary cause

If the disease always results from the factor, then we term the causative factor **"sufficient"**

Examples:

i. Polio virus is a necessary cause for paralytic polio but it is not a sufficient cause because only about one out of 50- 100 people who become in infected with poliovirus will develop paralytic polio.

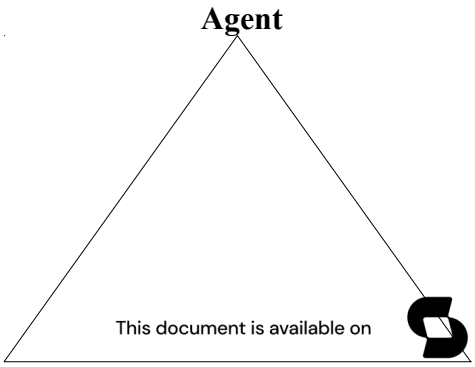
ii. The rabies virus is one of the very few risk factors which are both a necessary and a sufficient cause.

iii. Most single risk factors are neither necessary nor sufficient causes- for example, lung cancer occurs among non-smokers. (Smoking is not a necessary cause)

The concept of cause is not synonymous with a necessary or insufficient cause since many causes of disease are neither necessary nor insufficient.

The Epidemiologic Triad

The epidemiologic triangle or triad is the traditional model of infectious disease causation. It has three components: an external agent, a susceptible host, and an environment that brings the host and agent together. In this model, the environment influences the agent, the host, and the route of transmission of the agent from a source to the host.



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Host **Environment**

Agent factors

Agent originally referred to as an infectious microorganism – virus, bacterium, parasite, or other microbe. Generally, these agents must be present for disease to occur. That is they are necessary but not sufficient to cause disease.

As epidemiology has been applied to non-infectious conditions, the concept of agent in this model has been broadened to include chemical and physical causes of disease. This model does not work well for some non-infectious diseases, because it is not always clear whether a particular factor should be classified as an agent or as an environmental factor.

Host factors

Host factors are intrinsic factors that influence an individual's exposure, susceptibility, or response to a causative agent. Age, race, sex, socioeconomic status, and behaviors (smoking, drug abuse, lifestyle, sexual practices and contraception, eating habits) are just some of the many host factors which affect a person's likelihood of exposure. Age, genetic composition, nutritional and immunologic status, anatomic structure, presence of disease or medications, and psychological makeup are some of the host factors which affect a person's susceptibility and response to an agent.

Environmental factors

Environmental factors are extrinsic factors which affect the agent and the opportunity for exposure. Generally, environmental factors include physical factors such as geology, climate, and physical surroundings (e.g. a nursing home, hospital); biologic factors such as insects that transmit the agent; and socioeconomic factors such as crowding, sanitation, and the availability of health services

Agent, host and environment factors interrelate in a variety of complex ways to produce disease in humans. Their balance and interactions are different for different diseases. When we search for

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causal relationships, we must look at all three components and analyze their interactions to find practical and effective prevention and control measures.

Basic assumptions in Epidemiology

There are two basic assumptions in epidemiology

i. **Non- random distribution of diseases-** the distribution of disease in human population is not random on by chance. This explains why a certain individual or group gets a disease while another doesn't.
ii. Human disease has causal and preventive factors that can be identified through systematic investigation of different population or sub groups of individual within a population in different place and at different times.

Ecological approach to diseases

The occurrence and distribution of disease are studied against the back ground of various circumstances in man's total environment, physical, biological and social. This is an ecological approach to disease. The requirement that more than one factor be present for disease to develop is referred to as multiple causation or multifactorial etiology.

According to this concept, an agent of a disease be it a virus, bacteria or physical agent may be considered to be a necessary but the host and the environment must also be present for disease to develop.

Factors affecting the development of disease can be divided into 3 categories examples agent, host and environment.

Agent of disease (aetiological factor)

Nutrition elements- deficiency e.g. kwashiorkor; marasmus: excesses e.g. obesity.

Chemical agents- poison e.g. CO (carbon monoxide).

Physical agent- e.g. radiations.

Infectious agents-Protozoa e.g. amoeba.

Bacteria e.g. clostridium tetanus.

Fungi e.g. candida albacane.

Virus e.g. rota virus.

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Host factors (intrinsic factor)

Susceptible on response to agents such as:

Genetic	ethical group	immunological experience
Age	physiological experience	
Sex	human behavior	

Environmental factors (extrinsic)

Influence existence of the agent, exposure or susceptibility to agent such as:-

Biological environmental- which includes:

- Reservoirs infection
- Vectors that transmit disease
- Plants and animals.

Social environment

- Social economic status
- Political organizations.
- Culture

Physical environment- includes aspects of the environment such as:-

-Heat	-Water	-Gravity	-Chemical Agent
-Light	- Radiation	- Atmospheric Pressure	
-Air	-Frost	-Quakes.	

It is the interaction of these three set factors which determines whether or not disease develops.

Herd immunity

This is the level of immunity in community as a whole it indicates the decreased probability of a group on community developing an epidemic upon the introduction of an infectious agent, although these are a certain or of persons who are individually susceptible to the agent

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Nerd immunity is measured in terms of the proportion of immune or conversely of susceptible persons in a social group.

The presences of a large proportion of immune individuals in a community decrease the chances of contact between infected and susceptible persons. A very practical aspect of nerd immunity is that an entire population (100%) does not have to be immunized to prevent the occurrence of an epidemic.

Spectrum of disease

It is the sequence of events that occurs in the human organism from the time of exposure to the etiological agent to death. In infectious disease this spectrum is usually known as **gradient of infectious** diseases which reflects the sequence of manifestation of illness in the host reflecting his response to the infections at one extreme to death and to the other.

The frequency with which these different manifestations occur varies with the infectious diseases e.g. measles over 90% clinically apparent, Mumps about 60% clinically and Poliomyelitis over 90% not clinically.

Case definition

A case definition is a set of standard criteria for deciding whether a particular disease is related on another condition.

By using a standard case definition we ensure that every case is diagnosed in the same way regardless of when, where it occurred or who identified it.

We can then compare the number of cases of the disease that occurred in one time or place (e.g. Nairobi 1993) with the number that occurred at another time or another place (e.g. Nairobi 1995or Mombasa 1996)

A case definition consists of clinical criteria and sometimes limitation on time, place and person.

The clinical criteria usually include confirmatory laboratory tests, if available on combination of symptoms (subjective complaints) signs (objective physical findings) and other findings.

Example:

Clinical case definition:

UNCOMPLICATED MALARIA

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Fever or history of fever associated with symptoms such as nausea, vomiting and diarrhoea, headache, back pain, chills, myalgia, **where other infectious diseases have been excluded.**

WATERY DIARRHOEA

Diarrhoea i.e. 3 or more loose or watery stools over 24 hours with or without dehydration.

CHOLERA

Clinical case definition: Person aged over 5 years with severe dehydration from acute watery diarrhoea.

Person aged over 2 years with acute watery diarrhoea*in an area where there is a cholera outbreak.*

Confirmed case: Isolation of *Vibrio cholera O1* or *O139* from person with diarrhoea.

6. MEASLES

Clinical case definition: Person with:

fever, **and**

maculopapular (non-vesicular) rash, **and**

cough, coryza (runny nose) or conjunctivitis (red eyes).

Or any person in whom a medical officer suspects measles infection

7. MENINGITIS

Clinical case definition: Person with sudden onset of fever (>38.0 °C axillary)and one of the following:

- neck stiffness
- altered consciousness
- other meningeal sign

orpetechial or puerperal rash

In patients less than one year of age meningitis is suspected when fever is accompanied by a bulging fontanelle.

Confirmed case: Positive cerebro-spinal fluid antigen detection **or** positive culture.

Comment

Lab confirmation by all of the above methods is strongly recommended. A case definition may have several sets of criteria depending on how certain the diagnosis is e.g. during an outbreak of measles, we might classify a person with a fever or rash as being a suspect, probably a confirmed case of measles, depending as what additional case of measles was present. In other situation, we temporarily classify a case as suspect or probable until laboratory results are available. When we receive the lab reports we then re-classify the case either confirmed or not depending on the lab results.

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Note:

In the midst of the large outbreak of a disease caused by a known agent we may permanently classify some cases as suspect or probable because it is unnecessary and wasteful to run lab test.

WHY INVESTIGATE

- **Identify source of infection/s urgently**
- **Detect weakness in existing system**
- **To prevent mortality and morbidity**
- **To highlight public health needs**
- **Could result in policy changes**
- **To improve public health**
- **To be well informed about agent, host, and environment.**

PURPOSE: The course is designed to equip the learners with necessary knowledge, skills, and positive attitudes to be able to fulfill the below outlined objectives

Expected learning outcomes

By the end of the unit the learner should be able to:

i) Describe the frequency measures in epidemiology
ii) Identify the sources of epidemiological data

UNIT CONTENTS

Uses and purposes of epidemiology, terms used in epidemiology, frequency measures in epidemiology, rates, ratios and proportions, measures of morbidity, incidence, prevalence, relative risks, attributable risk, population attributable risks, measures of mortality, infant mortality rate, maternal mortality rate, prenatal mortality rate, neonatal mortality rate, case fatality rates, errors in measurement, sources of epidemiological data, census,

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USES AND PURPOSES OF EPIDEMIOLOGY

You may be able to make use of epidemiological information in the following ways:

viii. Diagnostic purpose

Epidemiological studies provide data for community diagnosis which encompasses the identification and detailed analysis of needs, demands, priorities and resources of a defined community.

This would serve as a basis for the planning and implementation of health intervention measures (this will usually be a cross sectional study as observations are made at specific time.)

ix. Identification of determinants of diseases

One of the most important purposes of epidemiological studies is the identification of determinants of diseases whose manipulation could lead to prevention and control of diseases.Hypothesis about disease causation may arise directly out of simple epidemiological description of diseases in terms of the kind of people affected by its (age, sex, occupation etc.) its geographical distribution and variation in its frequency of occurrence at different times e.g. pellagra and maize diet.

x. Surveillance

To produce enhanced surveillance for specific diseases e.g. cholera in order to be able to act quickly and to cut short any outbreak or pertussis (whooping cough). Surveillance is a useful monitor for the coverage achieved by an epidemiology programme.

xi. Investigation of an outbreak

To investigate an outbreak of a communicable disease, analyze the reason for it, plan a feasible remedy and carry it out and monitor the effects of the remedy on the outbreak.

xii. Observation of the natural history of diseases.

Knowledge of diseases natural history is essential to make prognosis or the likely outcome of a patient illness. This prognosis is the basis for rational decisions about therapy.

xiii. Classification of a disease

The epidemiological characteristics of a disease are an integral part of its basic description by means of which it is defined and recognized e.g.

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Gonorrhea classified as a STD based on its mode of transmission.

Shigellosis is grouped under diarrheal diseases based on its clinical manifestation of diarrhea and its fecal- oral route of transmission.

xiv. Planning

To plan effective health services. This particularly derived from community diagnosis for which effective services. Interventions and remedies all depend on accurate community data.

xv. Evaluation of methods of disease control

Any programme designed to prevent and control a disease must be accompanied by methods for accessing whether the measures are effective in reducing the frequency of the diseases. If the study is not effective then necessary alterations are done accordingly.e.g.

The effectiveness of health education on infant feeding to mothers attending anti-natal is evaluated by measuring decline in the incidence of marasmus and kwashiorkor among children and the change in knowledge, attitude and behavior of mothers. (This will be a longitudinal study as it involves the collection of information over a period of time and monitoring the change of health in a community.)

New terms used in Epidemiology II

Epidemics

Outbreaks of a disease affecting many people and usually caused by infections or in some cases dietary deficiency or poisoning

Disease Surveillance

It is an epidemiological practice by which the spread of disease is monitored in order to establish patterns of progression. The main role of disease surveillance is to predict, observe, and minimize the harm caused by outbreak, epidemic, and pandemic situations, as well as increase knowledge about which factors contribute to such circumstances

Epidemic

Epidemic occurs when new cases of a certain disease, in a given human population, and during a given period, substantially exceed what is expected based on recent experience (Any marked increase in the incidence or prevalence of disease).

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Pandemic

Pandemic is an epidemic of infectious disease that has spread through human populations across a large region; for instance multiple continents, or even worldwideeg HIV

Endemic

In epidemiology, an infection is said to be endemic in a population when that infection is maintained in the population without the need for external inputs(of a disease or condition regularly found among particular people or in a certain area).

Population at risk

This refers to the group of people who have the potential of getting the disease.

UNIT TWO: MEASURES OF MORBIDITY, MORTALITY AND ASSOCIATION

Frequency measures in Epidemiology

Ratio

It is a number devided by another number. A ratio expresses relationship between two items in form of x:y. These items may be either related or independent of each other. In the expression x:y, the counts of both items should be taken during the same interval. Example

The number of male and females in Kenya in the year 1998 were projected on the basis of the 1989 population census of Kenya. Males – 23, 630, 753

Females – 23, 674, 551

Total – 47, 305, 304

The ratio of Males to females can be obtained by 23, 630, 753÷23, 674, 551=0.998 apprx 1

Therefore the ratio can be expressed as 1:1

When the number of occurrence in a population are “a” and “b” then the ratio r is defined as

R= a or b or a:b or b:a
aa

To eliminate decimalpoints some time you may multiply by 10 so as to eliminate decimal points.

i.e $r=a/b\times10^k$

Where k is non negative number

Where k is 0 then 10 will be 1

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Where k is 1 then 10 will remain to be 10

E.g males 7 females 5

The ratio will be calculated as follows

$\frac{5}{7} \times 10^1 = 7.1$ females to 10 males

$\frac{5}{7} \times 10^2 = 71$ females to 100 males

$\frac{5}{7} \times 10^3 = 710$ females to 1000 males

Ratios are mostly expressed in this manner to enable easy comprehension.

Another example, There are 21,000 operations and 30 doctors in a hospital in a given year. The crude number given tells us about the work load and the staffing. Somebody may wish to consider the expansion of the hospital and how many Doctors may be required. He may want to know the ratio of the number of operations and the doctors. The ratio will be as follows

21,000÷30=700

This suggests that one Doctor is required for 700 operations. In this case k is 0

If we wanted to calculate the ratio to the related work for one Doctor then we will devide

30÷21,000= 0.0014 Doctors per operations

To eliminate the decimal we use 10^k (10^4)

30÷21,000 × 10000 = 14 Doctors per 10,000 operations

Proportion

A proportion is a specifictype of ratio in which the numerator is included in the denominator and the resultant value is expressed as a percentage. Example

The proportion of males in the total population in 1998 above is

$$\frac{\text{Males}}{\text{Males+Females}} \times 100$$

$$\frac{23630753}{} \times 100 = 49.95\%$$

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The formulae can be expressed as $P = \frac{a}{a+b}$

Proportion estimates the probability that an event might occur; if a proportion is more than 50% then the events is likely to occur.

Rates

A rate is the number of persons with a disease (or state, or event related to a disease) per unit of population per unit time.

To calculate rate, one requires not only the number of people (x) with a disease, but also the number (y) who do not have the disease and a denominator (x+y). The formula for the rate is given as follows

$$\text{Rate} = \frac{\text{Number of events in a specified period}}{\text{Pop. At rise of these events in a specified period}} \times K$$

The K (constant) is a round number greater than 1 and the most often used constants are 100, 1000, 10,000 or 100,000.

The advantage of rates over absolute numbers is the fact that they enable us to make comparison of occurrence of events (disease or deaths) in different population or in the same population at different times.

Example: The total population deaths and death rates in a hypothetical community between 1920-1980 is shown below

YEAR	TOTAL POP	DEATHS	DEATH RATE PER 1000
1920	10,000	600	60
1940	20,000	1200	60
1960	40,000	1800	45
1980	60,000	2000	33.3

Looking at the absolute number of deaths, one would think that this hypothetical community suffered more death in 1980, than in the preceding years. When we relate the absolute number of death to the total population at risk in the form of rate, it becomes evident that the population suffered less death than the previous years.

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Measures of Morbidity

Incidence

Defined as the number of New cases (or events) of a disease occurring in a population during a specified period of time. E.g Number of tuberculosis diagnosed in one year or number of number of pregnant mothers registered for the first time (new attendance) at the MCH clinics in one week.

Incidence measures new cases during a period of time, Incidence is not possible to measure directly because exact time of onset of illness is uncertain.

Incidence rate

Incidence rate is the measure of the rate at which new cases or events of disease condition occur in the population during a specified time.

$$\text{Incidence rate} = \frac{\text{No of new cases in a period of time}}{\text{Total population}} \times 100$$

Example. In a country with an estimated mid year population of 500, 000, 40 new cases of Kala-azar were reported in that year. What is the incidence rate? Relate this to every 100, 1000 and 100, 000 people

Inciden rate = 40÷500000= 0.00008
=0.008 pe 100
=0.08 per 1000

8.0 per 100,000

Uses

Incidence rates can be used to make statements about the probability or risk of a disease. They are compared among population groups with different exposures or attributes in order to measure the influence such factors may have on the occurrence of disease. Thus, estimates of Relative Risk are obtained when incidence rates are used to compare the frequency of diseases in different places.

Measures of Association

Relative Risk

Risk – is the probability that an event will occur e.g that an individual will become ill or die within a stated period of time or age, provided the individual continues being exposed to the

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risk factor. It is risky to be exposed to casual factors related to disease such as HIV, STD, T.B, Cancer e.t.c

Relative risk is also known as Risk Ratio

$$\text{Relative Risk} = \frac{\text{Incidence Rate (Exposed)}}{\text{Incidence Rate (Non exposed)}}$$

Incidence Rate Exposed refers to the rate of new cases of a particular disease diagnosed within the population already exposed to the causative factor that causes the disease.

Incidence rate non exposed (unexposed) refers to the rate of the new cases of a particular disease diagnosed on a population not exposed to the causative factor.

		Developed CHD			
		Yes	No	Totals	Incidence/1000
	Yes	84	2916	3000	28
Smokers	No	87	4913	5000	17.4

Incidence in smokers = 84/3000*1000= 28.0
Incidence in non smokers = 87/5000*1000 = 17.4
Relative Risk = 28.0/17.4= 1.61

Interpretation

1 = No association between exposure and disease

>1 = Positive association Exposed group has higher incidence than non-exposed group

<1 = Negatice association or protective effect- Non exposed group has higher incidence

A relative Risk of 1.0 or greater indicates an increased risk

A relative risk less than 1.0 indicates a decreased risk

Relative Risk Reduction

Relative risk measures how much the risk is reduced in the experimental group compared to a control group.

For example, if 60% of the control group died and 30% of the treated group died, the treatment would have a relative risk reduction of

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Attributable Risk or Absolute Risk (Population Attributable Risk- PAR)

It is the rate of the disease in the exposed individual that can be attributed to the exposure factor. In other words, It is excess risk of disease in total population attributable to exposure. It is the difference between incidence rate exposed-Incidence rate unexposed. It is an absolute measure of risk. It measures the impact of the removal of a certain factor may have on the incidence of disease

Example

Eating fish and stroke

Eating fish	Cases of stroke	Non cases of stroke	Total
Never	82 (a)	1549 (b)	1631 (a+b)
Almost daily	23 (c)	779 (d)	802 (c+d)
Total	105	2328	2433

Incidence in the exposed (*I*e): *a/a+b*_ 82/1,631 _ 0.0503, or 5.03 per 100
Incidence in the unexposed (*I*u): *c/c+d*_ 23/802 _ 0.0287, or 2.87 per 100
Incidence in both combined (*I*p): *a+c/(a _ b _ c _ d)* _ 105/2,433
0.0432, or 4.32 per 100

Relative Risk = Incidence Rate (Exposed) = a/a+b = 5.03 = 1.87
Incidence Rate (Non exposedc/c+d 2.87

Applying the formulas above to these data (and disregarding the fact that some members of the population may eat fish more than "never" and less than "almost daily") results in the following measures of attributable risk.

AR = $I_e - I_u$ = 5.03 – 2.87 = 2.16 per 100

AR% = (AR/I_e) 100 = (2.16/5.03) × 100 = 43%

PAR = $I_p - I_u$ = 4.32 – 2.87 = 1.45 per 100

PAR% = (PAR/ I_p) × 100 = (1.45/4.32) × = 33.6

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- Those who never eat fish have 1.75 times as much risk (higher incidence) as those who eatfish almost daily (*RR* = 1.75).
- If those who do not eat fish change their eating habits and begin to eat fish almost daily, their incidence of strokes will decrease by 2.16 per 100 individuals (*AR* = 2.16 per 100), which would represent a 43% reduction of their stroke incidence (*AR*% = 43%).
- A reduction of 1.45 new cases of stroke per 100 population (exposed and unexposed) is expected if everybody eats fish almost daily (*PAR* = 1.45 per 100). Such reduction represents a 33.6% reduction of the incidence in the population (*PAR*% = 33.6%).

Prevalence

It is the total number of cases at a particular point in a specified period. Prevalence rates measure the number of people in a population who have a disease at a given point in time. Total number of patients on the tuberculosis register who were being treated by The 1st of January will show the prevalence of diagnosed TB.

Prevalence measures all cases at one point in ztime; it is the number of instances of a given disease or other condition in a given population at a designated time. It means the extent at which a disease in a population exists in a given period.

The determinant of prevalence rate therefore is the incidence rate, cure rate, fatality level at a point in time or over a specified period.

UNIT THREE: DYNAMICS OF DISEASE TRANSMISSION

Contents;

- Communicable disease epidemiology
- Chain of disease transmission
- Methods of controlling communicable diseases
- Vaccine preventable diseases
- Expanded Program on Immunisation (EPI)

Definitions

i. **Communicable disease-** is an illness due to a specific infectious (biological) agent or its toxic products capable of being directly or indirectly transmitted from man to man, from animal to man, from animal to animal, or from the environment (through air, water, food e.t.c) to man

ii. **Infection-** is the entry and development or multiplication of an infectious agent in the body of man or animals. An infection does not always cause illness.

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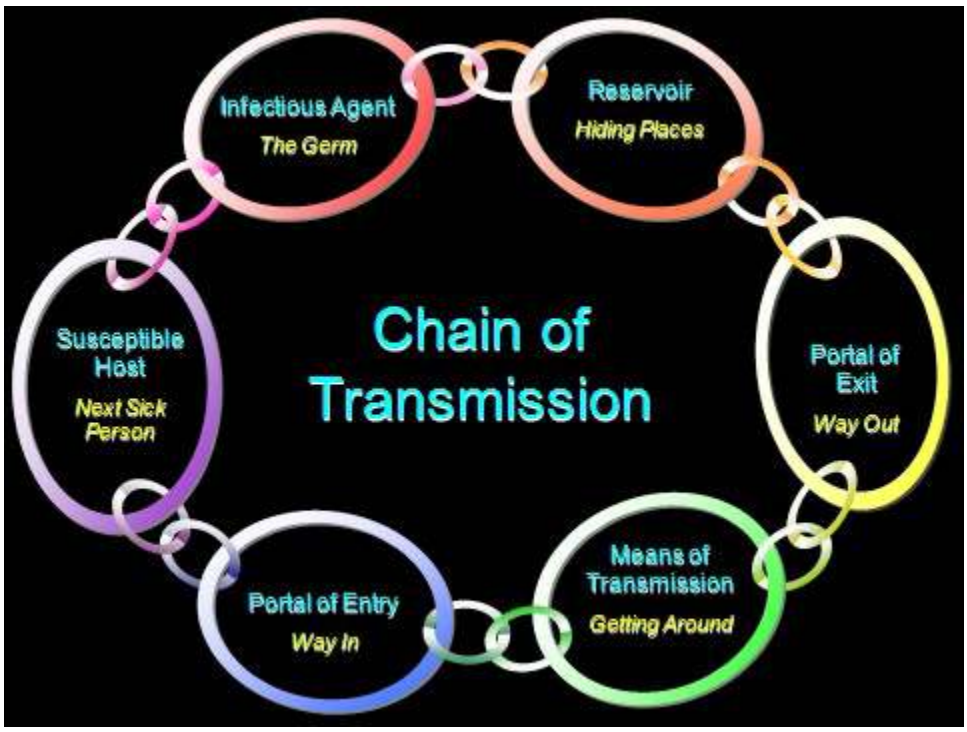
There are several levels of infection (Gradient of infection)0706534415

iii. **Fff**
iv. **Fff**
v. **Fff**
vi. **Fff**
vii. **Ff**
viii. **Fff**
ix. **Ff**
x. **Fff**
xi. **Fff**
xii. **Fff**
xiii. **Fff**
xiv.

CHAIN OF DISEASE TRANSMISSION

It is important to understand that infections don't just happen. A series of well defined circumstances or conditions need to be in place for an infection to occur. This is referred to as the chain of transmission. Interestingly, we do have some control in preventing infections as there are a number of strategies targeting each link on this chain.



METHODS OF CONTROLLING COMMUNICABLE DISEASES

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All of the infection prevention and control strategies are aimed at "breaking the chain" at one of the links of what we call the "Chain of Transmission".

Two ways of classifying communicable diseases

i. **Clinical classification** is based on the main clinical manifestations (symptoms and signs) of the disease.

   This way of classification is important in helping to treat the signs and symptoms that are common (shared by) individuals who suffer from different diseases.

   For example; *diarrheal diseases are characterized by passage of loose stool, febrile illnesses have fever, respiratory diseases are characterized by cough and shortness of braeth*

ii. **Epidemiological classification** is based on the main mode of transmission of the disease.

   The importance of this classification is that enables one to select prevention and control measures which are common to (shared by) communicable diseases in the same class, so as to interrupt the mode of transmission.

   Example of such classification include; ***waterborne diseases-*** *transmitted by ingestion of contaminated water,* ***foodborne diseases-*** *transmitted by the ingestion of contaminated food,* ***airborne diseases-*** *transmitted through the air* ***and vector-borne diseases-*** *transmitted by vectors such as mosquitoes and flies*

GENERAL APPROCAHES IN THE PREVENTION AND CONTROL OF COMMUNICABLE DISEASES

Prevention refers to measures that are applied to prevent (ensure that there is no occurrence) the occurrence of a disease while **control** refers to measures that are applied to prevent transmission after the disease has occurred.

This when applied to act on the chain of transmission by merging the six (6) links into 3 groups:

1. Infectious agents, the reservoir and portal of exit are put under 'reservoir'

   Measures here may include; ***diagnosis and treatment, screening, isolation***

2. Mode of transmission (in this respect we will look at measures that can be applied to interrupt transmission of infectious agents)

   Measures may include; ***water treatment, proper handling and storage of food, proper hygiene, sterilization of equipment***

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3. Portal of entry and the susceptible host are put under 'susceptible host'

   Such measures are aimed at protecting susceptible host either from becoming infected or from developing the clinical disease if they exposed to the infectious agent.

   Such measures may include; *vaccination, chemoprophylaxis, maintaining healthy lifestyle, limiting exposure to reservoirs of infection*

DISEASE PREVENTION

Definition; actions aimed at eradicating, eliminating, or minimizing the impact of disease and disability, if none of these are feasible, retarding the progress of disease and disability

Determinants of prevention

Successful prevention depends upon;

- ✓ A knowledge of causation
- ✓ Dynamics of transmission
- ✓ Identification of risk factors and risk groups
- ✓ Availability of prophylactic or early detection and treatment measures
- ✓ An organization for applying these measures to appropriate persons or groups
- ✓ Continuous evaluation of and development of procedures applied

Preventable causes of disease

Biological factors and **B**ehavioral factors

Environmental factors

Immunologic factors

Nutritional factors

Genetic factors

Services, **S**ocial factors and **S**piritual factors

LEVELS OF PREVENTION

The important tasks of health workers are:-

- ✓ Preventing people from getting sick.
- ✓ Preventing the sick people from getting worse.
- ✓ Preventing people from becoming disabled or chronically sick or dying.

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There are four important ways that health workers can prevent the development of diseases.

Primordial prevention

It consists of actions and measures that inhibit the emergence of risk factors in the form of environmental, economic, social and behavioral conditions and cultural patterns of living e.t.c

It is the prevention of the emergence or development of risk factors in countries or population groups in which they have not yet appeared eg *many adult health problems (such as obesity and hypertension) have their early origins in childhood, because this is the time when lifestyles are formed (for example smoking, eating patterns, physical exercise). In primordial prevention, efforts are directed towards discouraging children from adopting harmful lifestyles.*

The main intervention is through individual and mass education

Primary prevention

This is the level where intervention is aimed at healthy people, individuals or groups. Healthy people are prevented from becoming sick.

Measures are taken to promote optimum health or provide specific protection of targets groups against diseases and injury.

It may be defined as the action taken prior to the onset of disease, which removes the possibility that the disease will ever occur.

In order to carry effective primary prevention we must ask why healthy people become sick and also know who is "most at risk" of getting the disease. Some of the methods of primary prevention are:-

1. Immunization.
2. Provision/supply of safe water
3. Provision of adequate housing
4. Safe disposal of human excreta
5. Control of vectors of diseases
6. Protection against accidents and occupational hazards
7. Good nutrition, adequate clothing rest and recreation
8. Changing people's behavior through health education

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Secondary prevention

It is defined as action which halts the progress of a disease at its early stage and prevents complications

It attempts to arrest the disease process, restore health by seeking unrecognized disease and treating it before communicability of infectious diseases.

This involves detecting people (in the subclinical stage) who already have the disease as early as possible in order to cure the patient and prevent chronic disease and disability.

Measures are aimed at early diagnosis and prompt treatment to prevent the development of clinical disease and complications from the disease. This can be done by carrying a household survey or by telling people to visit the clinic to be examined by:

- Using laboratory tests to detect sub- clinical diseases
- Using clinical examination and lab tests to detect early clinical disease.

Tertiary prevention

It is defined as all the measures available to reduce or limit impairments and disabilities, and to promote the patient's adjustment to irremediable conditions

It is used when the disease process has advanced beyond its early stages.

This is the prevention of more disability and death. Measures at this level involve the treatment of the diseases (condition in its later stages and rehabilitation to optimize function thus preventing/ minimizing, impairment and disability.).

In a patient who cannot be cured, tertiary prevention has two aims;

- ✓ Treatment to prevent more disability and death
- ✓ New training and special education to help the patient to return to some useful work and life in the community (occupational therapy)

UNIT FOUR: DISEASE SURVEILLANCE AND OUTBREAK INVESTIGATIONS

- Definitions
- Types of epidemics
- Detection of epidemics
- Investigation of epidemics/epidemics
- Control of epidemics
- Definitions, types and purposes of surveillance
- Components of surveillance

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- Sources of surveillance data
- Steps in surveillance
- Strategy in disease surveillance
- Integrated Disease Surveillance and Response (IDSR)

UNIT FIVE: EPIDEMIOLOGICAL DATA AND STUDY DESIGNS

- Sources of epidemiological data
- Experimental studies, observational studies, descriptive studies, analytical studies, cross-sectional studies, longitudinal studies, case-control studies

UNIT SIX: PRINCIPLES OF DISEASE MEASUREMENTS

- Reliability, observer variation, variability, test bias and confounding, error and systematic error, repeatability, validity (content, concurrent, predictive and construct), sensitivity and specificity

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